RADANet: Road Augmented Deformable Attention Network for Road Extraction From Complex High-Resolution Remote-Sensing Images

Ling Dai, Guangyun Zhang, and Rongting Zhang

***Abstract*— Extracting roads from complex high-resolution remote sensing images to update road networks has become a recent research focus. How to apply the contextual spatial correlation and topological structure of the roads properly to improve the extraction accuracy becomes a challenge in the increasingly complex road environment. In this article, inspired by the prior knowledge of the road shape and the progress in deformable convolution, we proposed a road augmented deformable attention network (RADANet) to learn the long-range dependencies for specific road pixels. We developed a road augmentation module (RAM) to capture the semantic shape information of the road from four strip convolutions. Deformable attention module (DAM) combines the sparse sampling capability of deformable convolution with the spatial self-attention mechanism. The integration of RAM enables DAM to extract road features more specifically. Furthermore, RAM is placed behind the fourth stage of encoder, and DAM is placed between last four stages of encoder and decoder in RADANet to extract multiscale road semantic information. Comprehensive experiments on representative public datasets (DeepGlobe and CHN6-CUG road datasets) demonstrate that our RADANet achieves advanced results compared with the state-of-the-art methods.**

***Index Terms*— Deformable attention, high-resolution remote sensing images, road augmentation module (RAM), road extraction.**

I. Introduction

ROAD information plays the fundamental role in many applications, such as smart city construction, urban planning, vehicle navigation, and automatic drive [1], [2], [3]. To obtain up-to-date road information in large-scale areas, extracting road network from high-resolution remote-sensing images has become a promising method. High-resolution remote sensing imagery has become the main data source for extracting road regions and updating geospatial databases in real time [4]. Although road extraction from high-resolution satellite imagery recently gained considerable attention, this task remains challenging owing to irregular and complex road sections and structures [5]. Therefore, various methods for extracting roads from high-resolution satellite imagery have been proposed.

Traditional methods, such as texture progressive analysis [6] and mathematical morphology [7], are time-consuming and fraught with errors owing to human operators [8]. These approaches usually do not work for large-scale high-resolution imagery with complicated multiscale roads, because high-resolution imageries have distinct detail, which makes it tough to extract narrow and long-span roads.

With the rapid development of deep learning (DL), deep convolutional neural networks (DCNNs) have been proposed in road segmentation and extraction tasks and have acquired promising results.

However, road extraction from high-resolution remote sensing imagery is still full of challenges, for example, complicated urban environment largely affects road extraction because road has high similarity with other urban features. In order to avoid misclassifying other objects as road in areas of complex urban environments, the U-Net series [9], [10], [11] fully exploits road information with skip connection structure to collect multiscale information to improve performance. Similar to U-Net, LinkNet [12] also adapts encoder–decoder structure for semantic segmentation to improve the accuracy. Although DCNNs represented by U-Net and LinkNet have good performance in road semantic segmentation and extraction, difficulties still exist in road extraction.

1) Intraclass differences and interclass similarities of roads increase the difficulty of extraction. Diverse types of roads, such as urban roads, rural roads, railways, and so on, have obvious intraclass differences, and different objects, such as urban roads and urban buildings, have high interclass similarities. The two situations above will increase the difficulty for extracting the road context information [13]. Contextual information is helpful in extracting occlusions and tiny objects in a scene [14].
2) Some classic models have good performance for road extraction in relatively simple scenes, but they do not work in complicated scenes.

Current neural network models often fail to capture long-range dependencies in remote sensing imagery, resulting in limited extraction accuracy to a great extent. Accordingly, capturing long-range dependencies has become a focus in the field of

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DL society. Recent works [15], [16], [17], [18], [19], [20] have mainly focused on capturing long-range dependencies through some methods to expand the receptive field, such as multilayer convolution stacking, atrous convolution, feature pyramid, and multiscale convolution. However, these methods have different limitations, such as the poor performance of atrous convolution on small objects, and the stacking of multilayer convolutions will cause a surge in computation and low extraction efficiency. In addition, some of the state-of-the-art methods have integrated long-range dependencies with DL in diverse tasks. Liu et al. [21] proposed the self-constructing graph (SCG) module that learned a long-range dependency graph directly from the image data to improve semantic segmentation. Zhu et al. [22] proposed a Siamese global learning (Siam-GL) framework for semantic change detection, and the Siam-GL framework solved the problem that only local areas are considered in the existing framework. Inspired by the effectiveness of recurrent neural network (RNN) in modeling long-range dependency [23], [24], Li et al. [25] introduced RNN to exploit the relationship among different locations for high-resolution remote sensing image classification. Lei et al. [26] proposed a new DL network called selective nonlocal ResUNeXt++ (SNLRUX++) to establish long-range contextual dependencies for building extraction. Although the above methods performed well in capturing long-distance dependencies, they did not consider global information and local information at the same time. Recently, the transformer-based model has achieved remarkable results in computer vision tasks, and the self-attention mechanism, as the core part of the transformer, has acquired larger receptive field and contextual information by capturing the global information [27]. From a mathematical point of view, the self-attention mechanism learns the weights by taking the pairwise dot product of the input signals and then compares all the signals with the corresponding weights that are added together. Although the self-attention mechanism has a certain effect in capturing long-range dependencies, its limitations are also apparent. For remote sensing imagery, adjacent pixels on the spatial scale are highly similar and have strong dependencies. However, the self-attention mechanism emphasizes the semantic information of the global context and ignores the local information. The weighted sum operation of the global information will distract the attention weight and cause the range of interest to exceed the target range, resulting in invalid attention weight. So, semantic segmentation and extraction tasks are still challenging in identifying contextual semantic information. To better utilize contextual semantic information, deformable convolutions have been proposed, and deformable ConvNet (DCNv1) [28] learns an input-specific 2-D spatial offset to model geometric transformation. The modulated deformable ConvNet v2 (DCNv2) [29] adds a weight coefficient to distinguish whether the introduced area is an area of interest. Based on the deformable convolution and self-attention mechanism, multiscale deformable attention net (MDANet) combines the sparse spatial sampling strategy and the long-range relationship modeling capability [30]. The road class in the remote sensing imagery has special shape prior characteristics. Inspired by the characteristics of the road, we propose a road augmented deformable attention network (RADANet) to integrate the road shape prior into an end-to-end network. Among RADANet, the road augmentation module (RAM) has been designed to enhance road features, which incorporate the priory knowledge of the road shape. The proposed RADANet has U-shape encoder–decoder structure, and skip connections have been utilized for combining low-level features with high-level features simultaneously.

Comprehensive experiments demonstrate the superiority of our RADANet compared with several state-of-the-art methods. Compared with previous works, the contributions are summarized as follows.

1) An end-to-end DCNN framework called RADANet is proposed to extract complicated roads from high-resolution remote-sensing images. The deformable attention module (DAM) can capture low-level distinct information and high-level semantic information by inserting between encoders and decoders of the ResNet50.
2) The proposed RAM utilizes empirical knowledge that roads are almost long-span and narrow. Different from previous methods that extract road features with square convolution kernels, RAM designs strip convolutions in four different directions: horizontal, vertical, left diagonal, and right diagonal, which avoid to blur the boundary of the thin and narrow objects by traditional square convolution.
3) DAM applies spatial attention from local perspective instead of global input. We employ 1×1 convolution on the search space to a local scope of interrelated pixels for the reference pixel. Therefore, the learned offsets in DAM can adapt to the topological characteristics of roads, furnishing the reference pixel with spatial adaptability.

The remaining structure of this article is organized as follows. Section II introduces the related work of road extraction. In Section III, we present the detail of our proposed RADANet. In Section IV, implementation details are provided, and extensive experimental results are reported. Conclusion and discussion are presented in Section V.

II. Related Works

A. Road Segmentation

Road extraction from high-resolution remote-sensing images can be divided into two types: the traditional methods and the data-driven methods [31]. Traditional methods utilize hand-designed features and certain information about roads [32], [33], [34]. Zhang and Couloigner [35] introduced a number of descriptors of angular texture to identify the road segments with a fuzzy logic classifier. Wegner et al. [36] proposed a higher order conditional random field (CRF) model for extracting road network. Storvik et al. [37] described a Bayesian framework for road extraction, which was based on the multiscale features. He et al. [38] combined the results of boundaries estimation on the gray-level image and road-area extraction on the color image, presenting a color-based road detection algorithm. However, traditional approaches failed to achieve high accuracy and were inefficient and unsuitable for large-scale areas [39], particularly narrow road sections with high-width variance.

With rapid development of DL, the data-driven methods have achieved promising results. Some studies [40], [41], [42], [43], [44] extracted roads adopting CNN-based models. Mnih and Hinton [45] first extracted roads from high-resolution remote-sensing images with deep CNN on a graphics process. Saito et al. [46] extracted roads and buildings simultaneously from high-resolution remote-sensing images utilizing a single patch-based DCNN and strengthened its performance by postprocessing. Mei et al. [47] proposed a connectivity attention network (CoANet) to jointly learn the segmentation and pairwise dependencies for road extraction. Panboonyuen et al. [48] put forward a DCNN framework for road segmentation, the postprocessing of which adopted CRF to sharpen the extracted roads and used landscape metric to reduce misclassified road pixels. Zhang et al. [41] proposed a semantic segmentation neural network called residual neural unet (ResUnet), which integrated residual neural network (ResNet) and U-Net for road area extraction. These approaches acquired good road extraction results, but they failed to capture long-range dependencies of roads, resulting in insufficient contextual information during the process of road extraction.

B. Long-Range Dependencies in Networks

Some studies have designed networks that capture the long-range dependencies to improve performance in the task of roads semantic segmentation. Wang et al. [49] presented nonlocal operations as a generic family of building blocks for capturing long-range dependencies, which computed the response at a position as a weighted sum of the features at all positions. Wang et al. [50] introduced nonlocal LinkNet (NL-LinkNet) with differentiable nonlocal operations in order to model long-range dependencies. Zhu et al. [58] proposed global context-aware and batch-independent network (GCB-Net), which extracted complete and continuous road networks with global context features. Xie et al. [51] presented HsgNet for road extraction via learning global spatial information to establish long-distance context semantic associations. Lu et al. [52] proposed novel globally aware road detection network with multiscale residual learning (GAMS-Net) to capture the spatial context dependencies and interchannel dependencies. Xu et al. [53] developed an attention-based pyramid network to availably integrate spatial information with spectral information at different and same scales. Besides, relations between local features and global features were established. For the purpose of modeling the effective semantic interdependencies between each pixel pair of remote sensing data, Niu et al. [54] proposed a novel attention-based framework named hybrid multiple attention network (HMANet) to adaptively capture global dependencies. The approaches above aim to capture spatial context long-range correlations. However, our method improves the capacity for the acquisition of long-range dependencies. Among our approach, the direction and location of learned offsets sampling are in full accord with the topological structure of most roads in the high-resolution satellite imagery.

III. Methodology

A. Network Structure

In RADANet, the pretrained ResNet50 [59] has been employed as the encoder for its tremendous performance in feature extraction. As shown in Fig. 1, the whole framework adopts U-shaped structure. Inspired by fully convolutional network (FCN) [55], the decoder module uses bilinear interpolation to upsample the feature map of the last convolutional layer of the encoder, restoring it to the same size as the input image. Considering the special shape prior characteristics of the road class, RAM has been designed to enhance road features, which incorporate the priory knowledge of the road shape. The RAM has been inserted after Encoder_3. The reason why RAM is only embedded after Encoder_3 has been explained in ablation study in Section V. What calls for special attention is that the DAM has been embedded into four different stages of the pretrained ResNet50 for capturing multiscale context information of roads and intensifying topological correctness.

B. Self-Attention for Feature Intensifying

The self-attention mechanism can learn the similarity score between any position and all other positions to capture the long-range dependencies. The final feature map can be achieved by summing all positions with corresponding similarity score. For each feature vector $x_i \in \mathbb{R}^{N\times D}$ in the input image, N indicates the number of pixels in the feature, and D represents the feature channel. The self-attention module uses linear projection to compute three feature vectors: the query $q_i \in \mathbb{R}^{N\times D_q}$, the key $k_i \in \mathbb{R}^{N\times D_k}$, and the value $v_i \in \mathbb{R}^{N\times D_v}$

$$q_i = x_i W_q,\ k_i = x_i W_k,\ v_i = x_i W_v \tag{1}$$

where W_q, W_k, and W_v are the corresponding weight matrices. For the ith query at a given position and k_{all} at all positions, the similarity score between them can be calculated by dot product function

$$f(q_i, k_{\text{all}}) = q_i^T \cdot k_{\text{all}}. \tag{2}$$

The similarity scores are then normalized with the SoftMax function to obtain the attention weights

$$\text{Attention}(q_i, k_{\text{all}}) = \text{SoftMax}(f(q_i, k_{\text{all}}))\frac{e^{q_i^T \cdot k_{\text{all}}}}{\sum_{\forall \text{all}} e^{q_i^T} \cdot k_{\text{all}}}. \tag{3}$$

Then, the output can be expressed as

$$\text{output} = \sum_{m=1}^{M} W_m \left[\sum_{k\in\Omega q} \text{Attention}(q_i, k_{\text{all}}) \cdot W_m' v_i\right]. \tag{4}$$

This equation is represented as the sum of all the values v_i, which are weighted by corresponding attention weights. Let M represents the number of supporting search region, m indexes the attention head, Ωq indicates the supporting search area for the query q_i, and W_m and W_m' are of learnable weights.

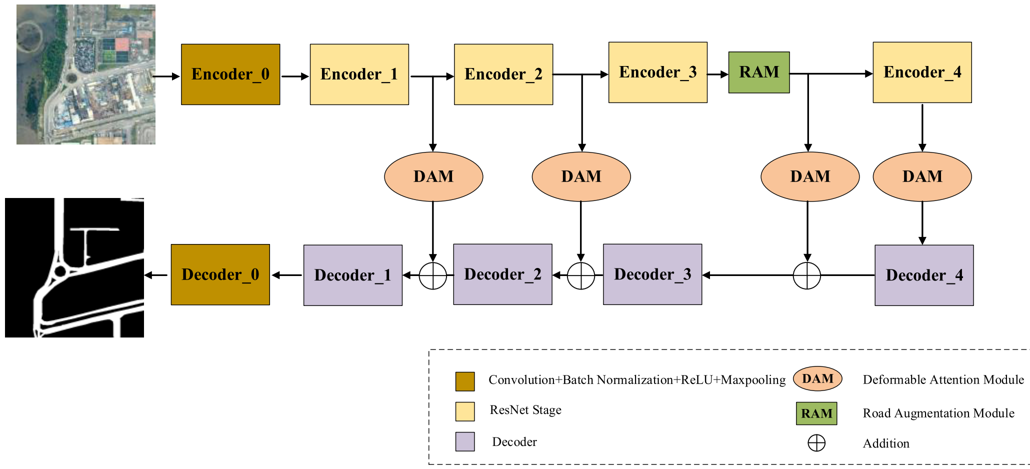

Fig. 1. Overall architecture of the proposed RADANet.

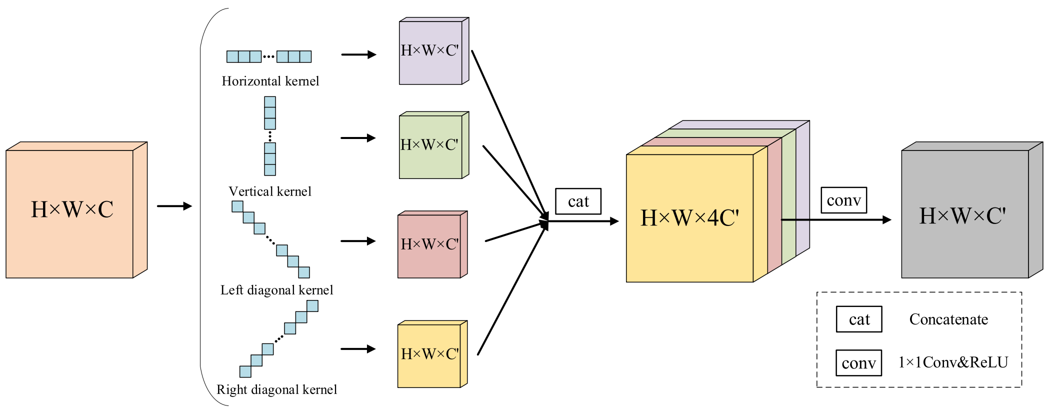

Fig. 2. Illustration of the RAM.

C. Road Augmentation

Compared with other types of features, the road class in the remote sensing imagery has special shape prior characteristics. Traditional convolutions often use square kernels, such as 3×3, 5×5, 7×7, and so on, because natural objects, such as buildings, vehicles, and even vegetation, are distributed in blocks. The road class with strip shape cannot be effectively extracted by traditional convolution kernel for its special shape prior characteristics. Therefore, strip convolution kernels are used to capture road long-range dependencies and prevent uncorrelated areas from disturbing the feature learning. Inspired by Sun et al. [60], we propose RAM. As shown in Fig. 2, RAM designs strip convolutions in four different directions: horizontal, vertical, left diagonal, and right diagonal, which avoid to blur the boundary of the thin and narrow objects by traditional square convolution. Let $X \in \mathbb{R}^{H\times W\times C}$ be the input of the RAM, where H, W, and C represent the height, width, and the number of channels. X is fed into four juxtaposed paths, and each path contains a convolution of one (not repeated) direction. Then, four output feature maps Y_h, Y_v, Y_l, and Y_r are obtained. The elementwise summation of the four feature maps $Y_h, Y_v, Y_l, Y_r \in \mathbb{R}^{H\times W\times C'}$ is performed to obtain $Y \in \mathbb{R}^{H\times W\times C'}$.

Let $w \in \mathbb{R}^{2m+1}$ denotes the convolution filter with the size $2m+1$. Filter w has the direction $D = (D_H, D_W)$, and $Y_D \in \mathbb{R}^{H\times W\times C'}$ is the output of convolution. The definition of such convolution can be given as

$$\begin{aligned} Y_D[i, j] &= (X * w)_D[i, j] \\ &= \sum_{t=-m}^{m} x[i + D_H t, j + D_W t] \cdot w[m-t] \end{aligned} \quad (5)$$

where $[i, j]$ represents the spatial position of any pixel, $X * w$ is the convolution operation, and D donates the direction vector of the convolution, which takes four values (0,1), (1,0), (1,1), and $(-1,1)$ for horizontal, vertical, left diagonal, and right diagonal, respectively [60]. To imitate the 3×3 convolution filter, we set $m = 4$ for w, so that each convolution kernel has nine parameters. Any position in the output feature map can be associated with multiple positions in the four directions in the input feature map.

D. Deformable Attention

Self-attention excels at capturing global long-range dependencies but poor at processing local contextual information. Deformable road attention is proposed to improve the generalization of self-attention for extracting road features. Inspired

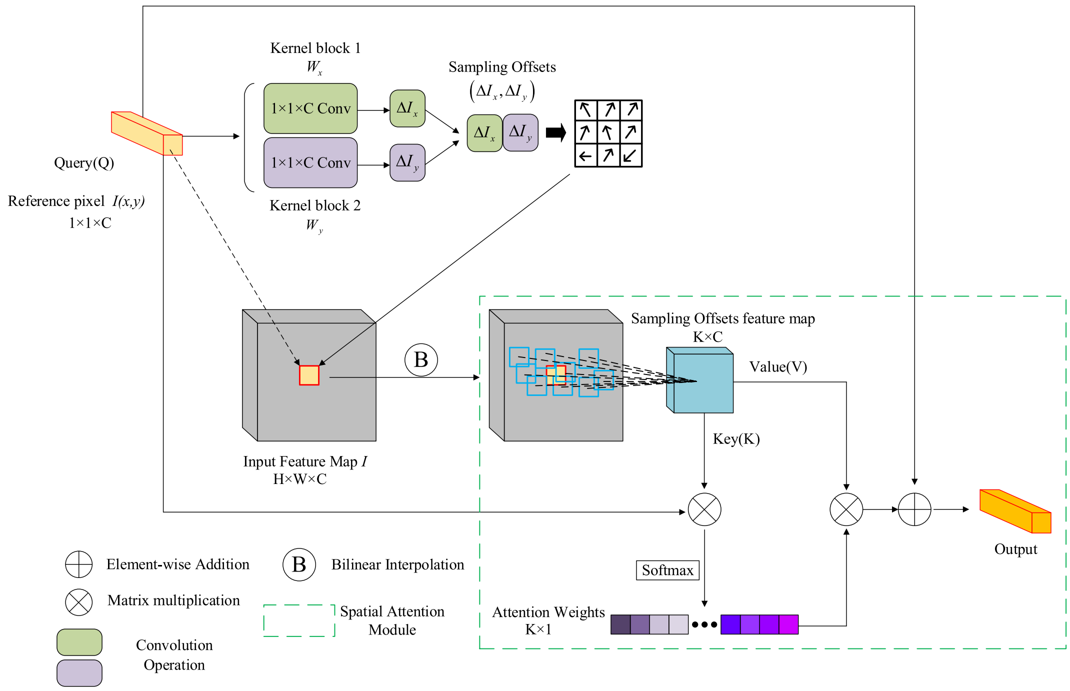

Fig. 3. Illustration of the proposed DAM. The extracted features can be recalibrated via inserting the spatial attention module after the output of the 1 × 1 convolution operation.

by Zhu et al. [29], [61], we set the feature map $I \in \mathbb{R}^{H\times W\times C}$, which H, W, and C denote the height, width, and the channel, respectively. Each pixel with location (x, y) in the feature map is represented as $I(x, y)$, and the 1×1 convolution operation in Fig. 3 is used to calculate the offset of each pixel for the input feature map. The offset can be represented as $(\Delta I_x, \Delta I_y)$, and the sampling process can be expressed as the following formula:

$$\left(\Delta I_x, \Delta I_y\right) = \left(I_Q(x, y)W_x^k, I_Q(x, y)W_y^k\right)_{k=1}^{K} \tag{6}$$

where K represents the total sampling pixels and k represents the arbitrarily sampled pixels. We assume $K = 9$ for the convenience of illustration. The weights (W_x^k, W_y^k) in 1×1 convolution sampling make sampling position be shifted to adjacent pixels.

The offsets are then added to the value of original sampling pixels for generating a new value of deflected pixels

$$I_\kappa(x_k, y_k) = I_Q\left((x + \Delta I_x), \left(y + \Delta I_y\right)\right). \tag{7}$$

Since the new value (x_k, y_k) of the position is noninteger, which cannot correspond to the actual pixels on the feature map, bilinear interpolation is required to confirm the most appropriate integer value by its few neighboring grid pixels

$$\begin{aligned} I_\kappa(x_k, y_k) &= \sum_{(x_{\text{int}}, y_{\text{int}})} G((x_{\text{int}}, y_{\text{int}}), (x_k, y_k)) \\ &\quad \cdot \sum_{(x_{\text{int}}, y_{\text{int}})} I_Q(x_{\text{int}}, y_{\text{int}}) \\ &= \sum_{(x_{\text{int}}, y_{\text{int}})} \max(0, 1 - |x_{\text{int}} - x_k|) \\ &\quad \cdot \sum_{(x_{\text{int}}, y_{\text{int}})} \max(0, 1 - |y_{\text{int}} - y_k|) \\ &\quad \cdot \sum_{(x_{\text{int}}, y_{\text{int}})} I_Q(x_{\text{int}}, y_{\text{int}}) \end{aligned} \tag{8}$$

where G represents bilinear interpolation in the x- and y-directions, (x_k, y_k) stands for arbitrary (fractional part) position, and $(x_{\text{int}}, y_{\text{int}})$ enumerates all integer locations in the feature map I_κ.

The 1×1 convolution operation can adapt to the special shape prior characteristics of the road automatically but does not consider the interaction between adjacent pixels. We introduce a spatial attention module to address this conundrum.

In spatial attention module, the Key (K) and the Value (V) are obtained from sampling feature map. We express $I_Q(x, y)$ as I_Q for convenience. The correlation between the specified pixel and all sampled pixels can be expressed as the dot product of I_Q and I_κ^T. I_κ^T represents transpose of I_κ. Besides, the softmax function is used to restrain the correlation between 0 and 1. The attention weights measure the interaction between the reference pixel and the κth sampled pixels

$$\text{Attention}_{Q\kappa} = \frac{\exp\left(I_Q I_\kappa^T\right)}{\sum_{\kappa=1}^{K} \exp\left(I_Q I_\kappa^T\right)}. \tag{9}$$

We add the attention weight $\text{Attention}_{Q\kappa}$ to all the corresponding sampled pixels for incorporating the priory knowledge of the road shape and retaining more road semantic information in a targeted manner. In order to maintain effective original road features, we integrate the context information of roads with the initial feature I_Q

$$I_{\text{output}} = \alpha \sum_{\kappa=1}^{K} \text{Attention}_{Q\kappa} I_V + I_Q \tag{10}$$

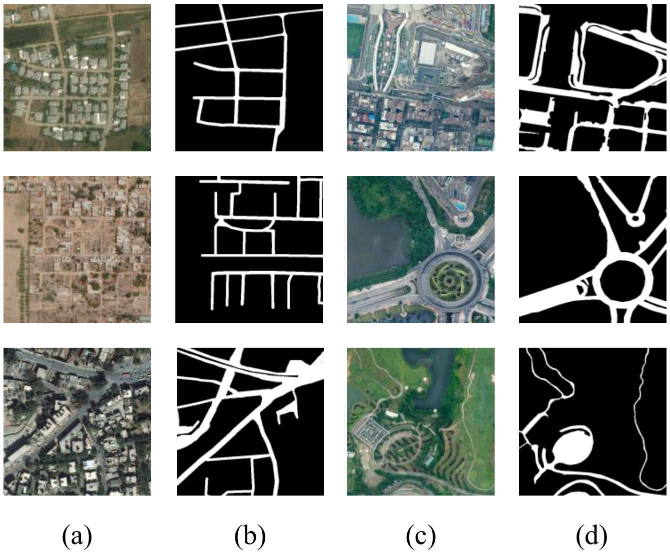

Fig. 4. Overview of the datasets and their groundtruth. (a) and (b) DeepGlobe roads dataset. (c) and (d) CHN6-CUG roads dataset.

where α represents a parameter of scale. The value of the parameter will rise from 0 during the training progress. The overall architectural of our proposed DAM is presented in Fig. 3.

IV. Experiments

A. Datasets

1) DeepGlobe Road Extraction Challenge Dataset [56]: As shown in Fig. 4, this dataset contains data with pixel-level annotations from Thailand, India, and Indonesia. The spatial resolution of imagery is 50 cm/pixel, and the image size is 1024 × 1024 pixels. The dataset contains 6226 images; following [57], we split it into 4976 images for training and 1250 for testing. The training dataset is augmented by creating crops of 512 × 512 pixels. In the following experiments, the dataset is called DeepGlobe dataset for short.

2) CHN6-CUG Roads Dataset [58]: This dataset is acquired from Google Earth. The images are selected from six representative cities in China: the Chaoyang area of Beijing, the Yangpu District of Shanghai, Wuhan city center, the Nanshan area of Shenzhen, the Shatin area of Hong Kong, and Macao. Each image has a size of 512 × 512 pixels and a resolution of 50 cm/pixel. The total 4511 images are split into 3608 images for train and 903 images for testing.

B. Evaluation Metrics

Extracting road from high-resolution imagery has been regarded as a binarized semantic segmentation problem. To evaluate the performance of our method for road extraction, we use $F1$-score, intersection over union (IoU) metrics, floating point operations (FLOPs), and number of parameters (Param). The pixelwise IoU score is defined as

$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}) \tag{11}$$

where TP means the number of pixels extracted as road, FP means the number of pixels which belong to other object but

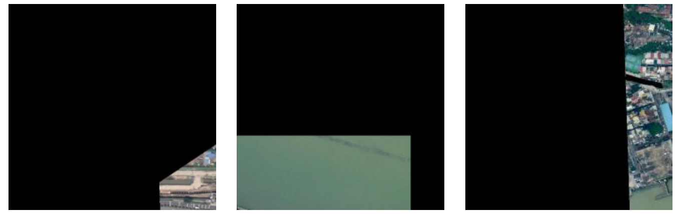

Fig. 5. Illustration of removed images in CHN6-CUG roads dataset.

extracted as road, and FN means the number of road pixels extracted as other object, and $F1$ score is given in (14)

$$\mathrm{precision} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \tag{12}$$

$$\mathrm{recall} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \tag{13}$$

$$F1 = 2 \times (\mathrm{precision} \times \mathrm{recall})/(\mathrm{precision}+\mathrm{recall}). \tag{14}$$

The FLOPs are often used to describe the computational complexity of a given model [66].

C. Implementation Details

In order to enhance the generalization of our RADANet, horizontal flipping and vertical flipping are employed for data augmentation. The adaptive moment estimation (Adam) [62] is used with the batch size of 2 and the initial learning rate is set to $1e^{-4}$. The epsilon and weight decay coefficients are set to $1e^{-8}$ and 0, respectively, and the decay rate of the first-order moment estimation is set to 0.9, and the decay rate of the second-order moment estimation is set to 0.999. The images are cropped to a fixed size of 512 × 512 for both datasets. Besides, our method is built on the basis of the PyTorch framework [63]. All the experiments are implemented on one NVIDIA RTX3060Ti graphics processing unit (GPU) with 6-GB memory. In CHN6-CUG roads dataset, we remove some useless images, which are the boundaries of large-scale satellite images, as shown in Fig. 5. Therefore, the dataset is left with 2835 images for training and 561 for testing, and the size of training images and test images is 512 × 512. Moreover, we reduce the DeepGlobe dataset and randomly select 2800 images for training and 500 images for testing, and the size of training images and test images is 512 × 512. Our approach can still get good results with limited computing resources. The number of training epoch for both datasets is 30 for convergence. Our RADANet is built up based on ResNet50, and we use the pretrained model of ResNet50 on the ImageNet dataset to speed up the training process.

D. Results

In this section, we draw a comparison between our RADANet and five state-of-the-art road extraction methods, including U-Net [9], Deeplabv3 [64], D-LinkNet [43], separable graph convolutional network (SGCN-Net) [65], and MDANet [30]. U-Net is one of the most popular road extraction frameworks. Researchers have further developed on its framework and proposed many improved models. Deeplabv3 extracts multiscale semantic information by stacking atrous convolutions. D-LinkNet is the winner of

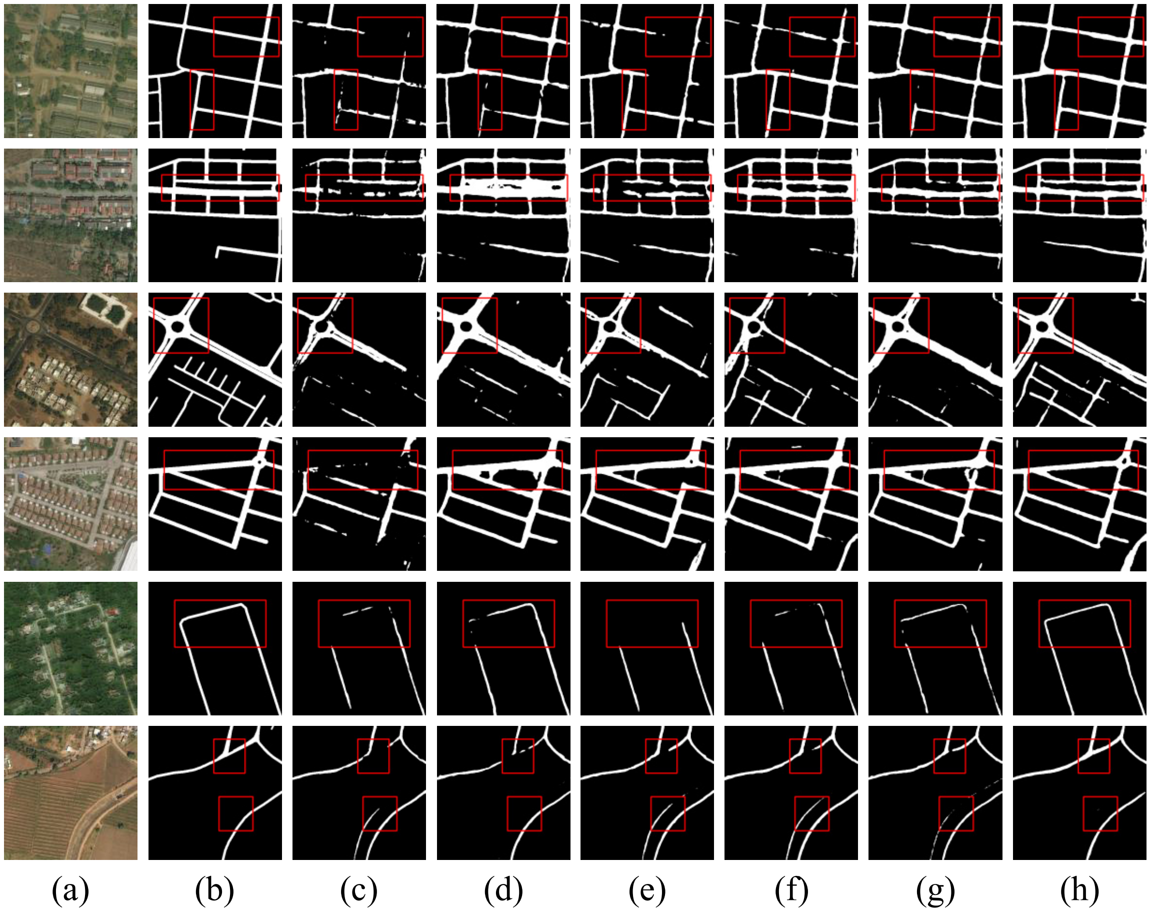

Fig. 6. Road extraction results using the DeepGlobe roads dataset. (a) Satellite imagery. (b) Ground truth. (c) U-Net. (d) Deeplabv3. (e) D-LinkNet. (f) SGCN-Net. (g) MDANet. (h) Ours.

the 2018 CVPR DeepGlobe Road Extraction Challenge, which improves LinkNet by adding atrous convolutions. SGCN-Net is a modified pixel-based graph convolutional network to capture global contextual road information in channel and spatial features. MDANet captures each pixel's adjacent configurable information via combining the sparse spatial sampling strategy and the long-range relationship modeling capability. As shown in Fig. 6, we make a comparison of all methods on DeepGlobe dataset and mark the key areas with red rectangles. In the first and fifth rows, some roads in the area are covered by vegetation and woods. In the second row, the road is not prominent, and the texture features of the road are similar to the background. Therefore, it is difficult to differentiate roads from other objects via integrating texture information with semantic information in the column of Fig. 6(c)–(e). In contrast, the relatively complete roads, which are extracted by our RADANet in the column of Fig. 6(h), are consistent with the ground truth. For other scenes, such as the third and fourth rows, the road intersection is more complicated than the roads in other rows. Despite the fuzzy road boundary in the sixth row, our RADANet consistently performs well.

As shown in Table I, compared with all other methods, our RADANet obtains the best Road IoU of 59.58% and $F1$ score of 74.67%. Specifically, compared with the worst-performing

TABLE I

Comparison of Our Proposed RADANet With Some Advanced Road Extraction Methods on the DeepGlobe Dataset. The Best Results Are Highlighted in Boldface

Model	Road IoU (%)	F1 Score (%)	Param (M)	FLOPs (M)
U-Net	48.87	65.99	34.53	5.24E+05
DeeplabV3	53.57	69.77	40.35	1.38E+05
D-LinkNet	53.12	69.68	217.65	2.4E+05
SGCN-Net	52.02	68.44	42.73	1.25E+05
MDANet	52.15	68.55	**31.9**	**8.86E+04**
ours	**58.58**	**73.67**	73.85	2.12E+05

U-Net, our method makes an outstanding improvement in IoU of 10.71% and $F1$ score of 8.68%; besides, the proposed method is better than the second best performing deeplabv3 by 6.01% and 4.9%. Roads in DeepGlobe dataset almost belong to rural roads. The texture features of roads are similar to the background in rural areas, and roads are covered by vegetation and shadow. Nonetheless, our RADANet still achieves good results. Compared with the DeepGlobe dataset, the roads contained in the CHN6-CUG dataset are more complex, including urban road, forest road, country road,

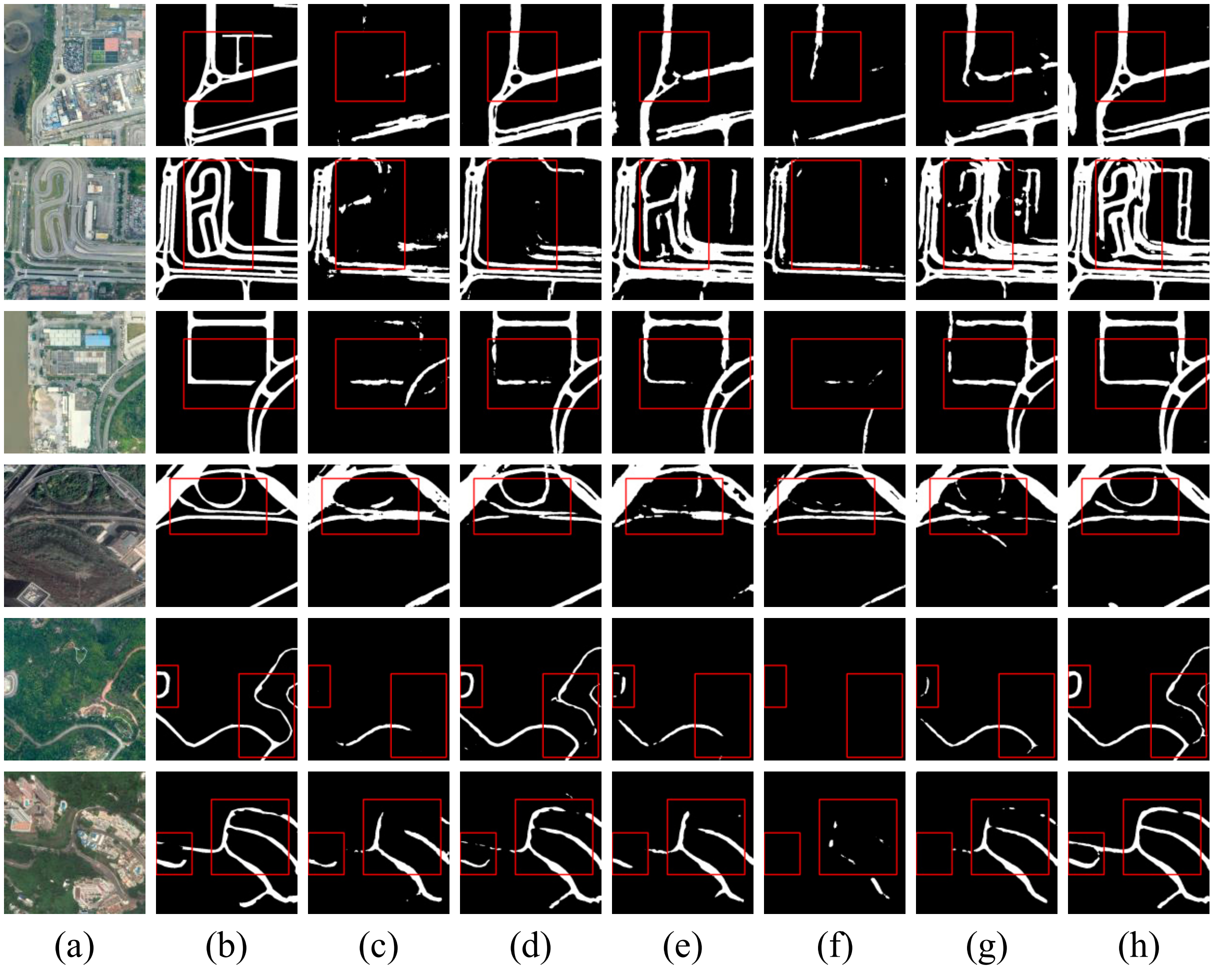

Fig. 7. Road extraction results using the CHN6-CUG roads dataset. (a) Satellite imagery. (b) Ground truth. (c) U-Net. (d) Deeplabv3. (e) D-LinkNet. (f) SGCN-Net. (g) MDANet. (h) Ours.

factories, mines road, built-up road, railway crossing, overpass, viaduct, and intersection [58].

In order to further verify the robustness and generalization of our RADANet in complex road scenes, we also conduct experiments on the CHN6-CUG dataset. As shown in Fig. 7, the first four rows contain complex road scenes in the city. The fifth and sixth rows have road scenes of mountain path and mountain highway, respectively, and some part of them are covered by trees. Our RADANet still has advantages over other methods [see Fig. 7(c)–(g)] when extracting complicated and diverse roads. Only Deeplabv3's performance is close to our model, indicating that the multiscale context extracted via atrous convolution is relative effective.

As shown in Table II, our RADANet has superior quantitative results on the CHN6-CUG dataset, which obtains an IoU score of 60.43% and an $F1$ score of 75.34%. Compared with the SGCN-Net, our method makes an outstanding improvement in IoU of 16.17% and $F1$ score of 13.98%; besides, the proposed RADANet improves the second-best Deeplabv3 by 6.66% and 5.4%. Params and FLOPs are also considered. These two indicators represent the number of parameters and calculation of the model, respectively, and measure the complexity of the model. The Params and FLOPs of our RADANet are 73.85(M) and 2.12E + 05(M), respectively. Compared with the other five state-of-the-art methods, the Params and FLOPs of our model are relatively large. Only D-LinkNet has larger Params and FLOPs than ours. However, considering that our RADANet has excellent performance in IoU and $F1$ score, large Params and FLOPs are not unacceptable.

TABLE II

QUANTITATIVE COMPARISON OF OUR PROPOSED RADANET WITH SOME ADVANCED ROAD EXTRACTION METHODS ON THE CHN6-CUG DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLDFACE

Model	Road IoU (%)	F1 Score (%)	Param (M)	FLOPs (M)
U-Net	45.05	62.11	34.53	5.24E+05
DeeplabV3	53.77	69.94	40.35	1.38E+05
D-LinkNet	52.03	68.45	217.65	2.4E+05
SGCN-Net	44.26	61.36	42.73	1.25E+05
MDANet	53.16	69.69	**31.9**	**8.86E+04**
ours	**60.43**	**75.34**	73.85	2.12E+05

TABLE III
ABLATION STUDY FOR COMPARING THE EFFECT OF DIFFERENT BATCH SIZE VALUES ON THE ROAD IOU AND $F1$ SCORE UNDER THE CONDITION THAT THE INPUT SIZE IS 512×512

Model Name	DeepGlobe Roads Dataset		CHN6-CUG Road Dataset		batch size
	Road IoU (%)	F1 Score	Road IoU (%)	F1 Score	
U-Net	49.73	65.79	46.87	64.02	4
	54.68	70.70	50.37	68.63	6
	55.52	71.02	52.64	70.88	8
DeeplabV3	55.92	71.19	53.99	69.41	4
	56.31	71.24	55.73	72.67	6
	56.49	71.33	56.26	72.77	8
D-LinkNet	55.52	71.41	51.98	68.93	4
	56.61	72.29	52.14	69.39	6
	57.21	72.44	53.47	70.68	8
SGCN-Net	52.32	68.06	45.73	62.42	4
	52.19	69.34	48.01	65.84	6
	53.05	69.79	49.19	66.95	8
MDANet	54.15	70.55	53.17	69.42	4
	55.19	70.95	55.02	71.56	6
	55.37	70.98	55.83	71.72	8
ours	**59.67**	**74.98**	**60.51**	**75.32**	4
	59.94	**75.23**	**60.57**	**75.41**	6
	60.46	**75.42**	**61.73**	**76.34**	8

V. DISCUSSION

A. Ablation Study I

The values of batch size and input size are hyperparameters in the experiment. In order to verify the influence of these two hyperparameters on the accuracy of the model, we conduct the ablation study. The setting of these hyperparameters usually depends on the hardware facilities of the computer. Some large laboratories pursue the ultimate accuracy of the model because they have excellent hardware conditions. The training results of deep neural networks are related to the GPU memory. The larger the GPU memory, the more images that can be processed at one time, and the higher the accuracy of the model may be. However, it should be noted that the improvement of computer hardware capabilities cannot change the merits of the model itself. It can only be used as an auxiliary means to improve the training accuracy of the model. In the above experiments, the results of our experiments are based on 6-GB GPU memory, and the value of batch size can only be set to 2. In ablation study, all the experiments are implemented on one NVIDIA RTX3090 GPU with 24-GB memory. The values of the batch size and the input size can be set to 2, 4, 6, and 8 and 352, 480, and 512, respectively. As shown in Table III, in general, as the value of batch size increases, the values of Road IoU and $F1$ score of each model almost show an upward trend. Our model still maintains its superiority over other models. Compared with the results in Tables I and II, when the value of batch size is set to 8, the road IoU and $F1$ score of our model in the DeepGlobe dataset are improved by 1.88% and 1.75%; the road IoU and $F1$ score in the CHN6-CUG dataset are improved by 1.3% and 1%, respectively.

Besides, we discuss the influence of different input sizes on the results with a fixed value of batch size. As represented in Table IV, when the value of input size increases from 352 to 480, the road IoU and $F1$ score of each model are improved. The results of our model are still ahead of other models. Combining the results of Tables I and II, when the input size increases from 352 to 512, the two indicators of our model on the DeepGlobe dataset are improved by 4.32% and 3.5% and are improved by 3.9% and 3.15%, respectively, on the CHN6-CUG dataset.

B. Ablation Study II

As shown in Table V, we discuss the effects of different position for RAM. In Section IV, the RAM has been inserted after Encoder_3. When RAM is inserted after other position, such as Encoder_1, Encoder_2, and Encoder_4, RADANet achieves relatively poor performance. However, it still outperforms the other five state-of-the-art models. We also report results under the condition that RAM is inserted after the encoders from Encoder_1 to Encoder_4. As shown in No. 5, it is bad for improving performance and it brings high computational cost. Therefore, we only insert RAM after Encoder_3. In addition, we conduct the comparative experiments between RADANet with and without RAM. For convenience, we discuss the No. 3 in Table V, namely, RAM is inserted after Encoder_3. As shown in Table VI, the IoU of RADANet without RAM in the DeepGlobe dataset decreases by 4.37%, and that in the CHN6-CUG dataset decreases by 2.96%. Therefore, RAM has an important impact on model accuracy.

TABLE IV

ABLATION STUDY FOR COMPARING THE EFFECT OF DIFFERENT INPUT SIZE VALUES ON THE ROAD IOU AND $F1$ SCORE UNDER THE CONDITION THAT THE BATCH SIZE IS 2

Model Name	DeepGlobe Roads Dataset		CHN6-CUG Road Dataset		input size
	Road IoU (%)	F1 Score	Road IoU (%)	F1 Score	
U-Net	46.20	63.21	34.35	51.14	352×352
	48.79	66.13	41.29	58.06	480×480
DeeplabV3	45.25	62.26	51.36	67.86	352×352
	48.44	64.51	52.04	68.47	480×480
D-LinkNet	45.95	62.97	52.71	69.03	352×352
	48.11	64.96	52.89	69.21	480×480
SGCN-Net	42.73	59.88	40.53	57.68	352×352
	45.77	62.93	42.71	59.86	480×480
MDANet	48.40	64.62	48.11	64.96	352×352
	50.04	66.57	48.77	65.09	480×480
ours	**54.26**	**70.17**	**56.53**	**72.19**	352×352
	57.57	**73.06**	**60.04**	**75.03**	480×480

TABLE V

ABLATION STUDY FOR THE PROPOSED RAM WITH DIFFERENT POSITIONS IN ENCODER UNDER THE CONDITION THAT THE BATCH SIZE IS 2 AND THE INPUT SIZE IS 512 × 512. "E_1," "E_2," "E_3," AND "E_4" REPRESENT ENCODER_1, ENCODER_2, ENCODER_3, AND ENCODER_4, RESPECTIVELY

No.	E_1	E_2	E_3	E_4	DeepGlobe IoU (%)	DeepGlobe F1 Score (%)	CHN6-CUG IoU (%)	CHN6-CUG F1 Score (%)
1	√				57.03	72.33	58.62	73.92
2		√			55.78	71.62	58.37	73.71
3			√		**59.58**	**74.67**	**60.43**	**75.34**
4				√	57.81	73.02	58.43	73.76
5	√	√	√	√	58.71	73.98	59.29	74.45

TABLE VI

ABLATION STUDY FOR THE PROPOSED RADANET WITH AND WITHOUT RAM. THE RADANET WITH RAM REFERS TO NO. 3 IN TABLE V

	DeepGlobe		CHN6-CUG	
	IoU (%)	F1 Score (%)	IoU (%)	F1 Score (%)
with RAM	**59.58**	**74.67**	**60.43**	**75.34**
without RAM	55.21	70.18	57.47	72.23

C. Ablation Study III

In order to explore the influence of the presence and number of DAMs on our model, we conduct the following ablation experiments on the CHN6-CUG roads dataset to verify the excellent performance brought by the proposed DAM at different positions of the ResNet50. DAM-1, DAM-2, DAM-3, and DAM-4 refer to the Encoder_1, Encoder_2, Encoder_3, and Encoder_4 inserted with DAM, respectively, while DAM-0 refers to no DAM inserted in ResNet50 and DAM-5 refers to all encoders inserted with DAM (except Encoder_0). Noted that we do not consider the effects of Encoder_0 due to its simple structure. As shown in Table VII, the results are highly related to the inserted position for DAM. The hyperparameter settings of the experiment are consistent with those in Table II. Specifically, compared with DAM-0, DAM-5 improves the IoU and $F1$ by 10.46% and 7.07%.

TABLE VII

ABLATION STUDY FOR THE PROPOSED DAM WITH DIFFERENT POSITIONS AND NUMBERS

DAM	IoU (%)	F1 Score (%)
DAM-0	49.97	68.27
DAM-1	51.71	69.43
DAM-2	54.19	71.27
DAM-3	55.84	72.01
DAM-4	56.27	72.08
DAM-5	**60.43**	**75.34**

D. Ablation Study IV

In this section, we discuss the influence of the number of training samples on the experimental results. In the above experiment, we only select a small part as training sample for DeepGlobe dataset and only 2800 images with the size of 512 × 512. In order to further illustrate the superiority of our model, we increase the number of training samples to 16 800 images with the size of 512 × 512 and conduct the experiment under the condition that the batch size is 8 and the GPU memory is 24G. For the CHN6-CUG dataset, although we have used almost all training samples, we combine the original training samples and test samples to form a new set of training samples, and the number of training samples are increased to 3396 images with the size of 512 × 512. As shown in Table VIII, the performance of all models has been improved to vary degrees with sufficient training data and computing power, and our model is still superior to other models. Compared with the results in Table III, road IoU and $F1$ score of our model in the DeepGlobe dataset are improved by 3.76% and 3.3%; the road IoU and $F1$ score in

TABLE VIII

ABLATION STUDY FOR DIFFERENT MODELS ON THE ROAD IoU AND $F1$ SCORE UNDER THE CONDITION THAT THE BATCH SIZE IS 8 AND THE GPU MEMORY IS 24G. THE TRAINING SAMPLE OF DEEPGLOBE DATASET AND CHN6-CUG DATASET ARE 16 800 AND 3396

Model	DeepGlobe		CHN6-CUG	
	IoU (%)	F1 Score (%)	IoU (%)	F1 Score (%)
U-Net	60.37	75.16	52.67	70.91
DeeplabV3	60.81	75.32	56.92	73.15
D-LinkNet	62.09	77.01	55.07	71.85
SGCN-Net	57.49	73.77	49.65	67.42
MDANet	60.21	75.19	56.08	71.93
ours	**64.22**	**78.72**	**62.57**	**77.15**

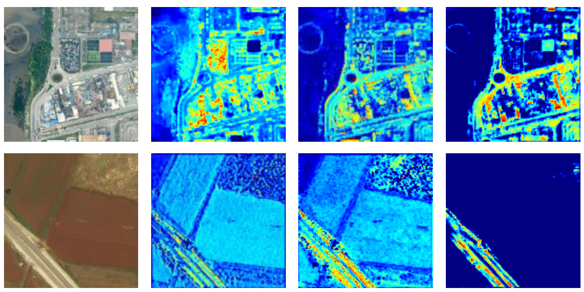

Fig. 8. Visualization of features for No.1 in Table V. (a) Input images, (b) before adding RAM, (c) after adding RAM, and (d) after adding DA in (c).

the CHN6-CUG dataset are improved by 0.84% and 0.81%, respectively. Therefore, our model can get good results with a small number of training samples, and when the training samples are increased, the performance of our model has improved significantly, which is better than other models. This further demonstrates the superiority of the model.

E. Feature Visualization for Roads

In this section, in order to demonstrate how RADANet extracts road features, we randomly select two images for feature visualization. We take No. 1 in Table V as an example for convenience, because as the network goes deeper, the extracted features will become more abstract. As shown in Fig. 8, after adding the RAM, the road information is enhanced, and after adding the DA, the edges of the learned road features are more concentrated and clearer, and the influence of redundant information on road extraction is reduced.

VI. CONCLUSION

In this article, we propose a RADANet for road extraction from complex high-resolution remote-sensing imagery. In general, we adopt the U-shape encoder–decoder structure to learn road features. Taking into account the special shape prior characteristics of the road class, the RAM has been designed to enhance the semantic features of roads, which incorporate the priory knowledge of the road shape. The DAM has been proposed by incorporating offset into the self-attention mechanism. DAM can effectively capture long-range dependencies between road pixels. Furthermore, the RADANet can capture multiscale road semantic information by embedding DAM between each layer of encoder and decoder, and the network can achieve the best performance when the RAM is inserted after Encoder_3. We conduct experiments on two public datasets (DeepGlobe roads dataset and CHN6-CUG road dataset), and the results show that our RADANet performs competitively in the face of complex road. Besides, our RADANet consistently outperforms other five state-of-the-art models under different conditions in ablation experiments.

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